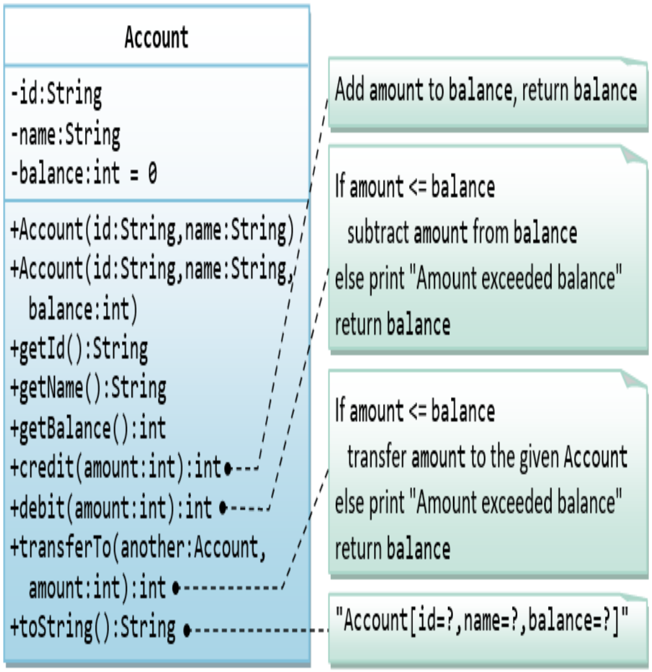
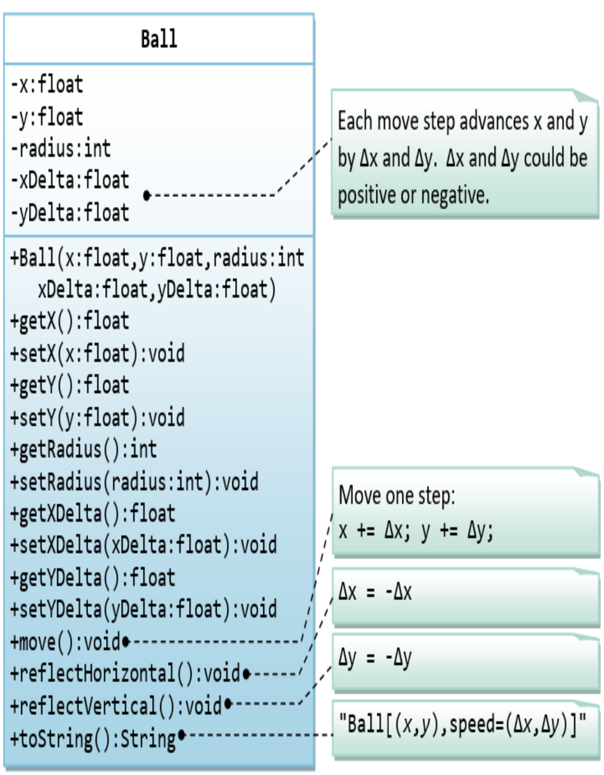
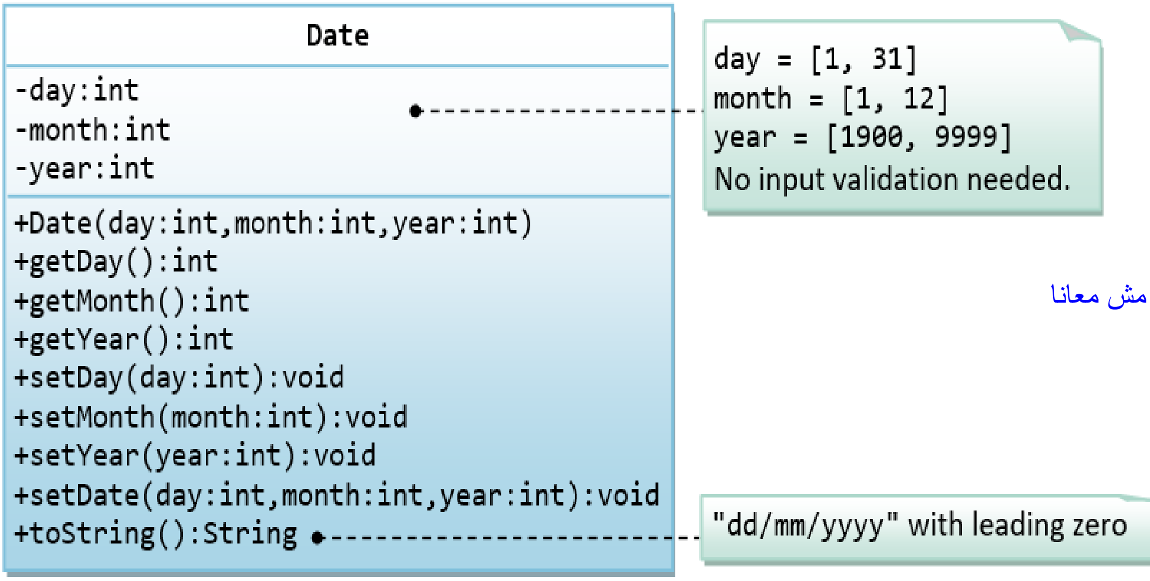
A class called Account, which models a bank account of a customer, is designed as shown in the following class diagram. The methods credit(amount) and debit(amount) add or subtract the given amount to the balance. The method transferTo(anotherAccount, amount) transfers the given amount from this Account to the given anotherAccount. Write the Account class.



A class called Ball, which models a bouncing ball, is designed as shown in the following class diagram. It contains its radius, x and y position. Each move-step advances the x and y by delta-x and delta-y, respectively. delta-x and delta-y could be positive or negative. The reflectHorizontal() and reflectVertical() methods could be used to bounce the ball off the walls. Write the Ball class. Study the test driver on how the ball bounces.



A class called Date, which models a calendar date, is designed as shown in the following class diagram. Write the Date class.



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A class called Time, which models a time instance, is designed as shown in the following class diagram. The methods nextSecond() and previousSecond() shall advance or rewind this instance by one second, and return this instance, so as to support chaining operation such as t1.nextSecond().nextSecond(). Write the Time class.

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